Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 2

Date: _____ Name: _____

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**30**
Part 2	**25**
CW	**5**
Marks Obtained	

Section A:

MCQ: **(5x1=5 marks)**

For each question 1-5, four options are given. Circle the correct option i, ii, iii or iv.

1. When a large file is sent across a network, it is split into smaller units called packets. What is the typical size of each packet?

i. 1 Byte **ii.** 64 KiB

iii. 10 GiB **iv.** 500 MiB

2. In packet switching, which network device reads the destination IP address and directs each packet along the best route?

i. Monitor **ii.** Keyboard

iii. Router **iv.** Scanner

3. A standard USB cable consists of a four-wire shielded cable. What are the 4 wires used for?

i. 4 wires for power only **ii.** 2 wires for power and 2 wires for data

iii. 4 wires for sound only **iv.** 1 wire for power and 3 wires for video

4. What is the maximum recommended cable length for a standard USB connection before signal hub repeaters are needed?

i. 5 metres **ii.** 500 metres

iii. 5 kilometres **iv.** 100 kilometres

5. Which transmission method sends data bits one after another (one bit at a time) down a single channel or wire?

i. Parallel data transmission **ii.** Serial data transmission

iii. Skewed data transmission **iv.** Analogue transmission

Section B:

Matching Terms: **(5x1=5 marks)**

6. Match each term in Column A with its correct description in Column B:

Column A (Term)	Column B (Description)	Answer
i. Payload	**A.** Marks the end of the packet and contains error-checking data (CRC).	**i →** ____
ii. Packet Trailer	**B.** Bits arrive out of synchronization across parallel wires over long distances.	**ii →** ____
iii. Bit Skew	**C.** The actual user data being carried inside the packet (about 64 KiB).	**iii →** ____
iv. Crosstalk	**D.** Computer automatically detects a USB device and loads its driver.	**iv →** ____
v. Plug and Play	**E.** Electromagnetic interference between adjacent wires in parallel cables.	**v →** ____

Section C:

Direction of Data Transmission: **(3x2=6 marks)**

7. Complete the table below by stating the direction of data flow and one real-life example:

Mode	How Data Flows (1 mark each)	Real-Life Example (1 mark each)
i. Simplex		
ii. Half-Duplex		
iii. Full-Duplex		

Section D:

Serial vs. Parallel Transmission: (4x1=4 marks)

8. Tick (✓) whether each statement describes Serial or Parallel data transmission:

No.	Statement / Feature	Serial (✓)	Parallel (✓)
i.	Bits are sent one after another along a single wire or channel.		
ii.	Multiple bits (e.g., 8 bits / 1 byte) are sent simultaneously across several wires.		
iii.	Most reliable over long distances because there is zero risk of bit skew.		
iv.	Used inside a computer motherboard (internal buses) over very short distances.		

Section E:

Short Questions (Packet Switching & USB): (2+3=5 marks)

9. State any TWO items of information stored inside a Packet Header: (2 marks)

i.

_____

ii.

_____

10. State TWO advantages and ONE limitation of using the Universal Serial Bus (USB) interface: (3 marks)

i. Advantage 1:

_____

ii. Advantage 2:

_____

iii. Limitation:

_____